Policy Gradient (continue)

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CS 1840 Fall 2026

Announcements

HW4 is out and due 11/12 (this Thursday)!

Midterm corrections for half points back
(Check the ed discussion post for details)!

Check the course website for Office Hours:
https://harvard-cs-1840.github.io/2026/

Warm Up: Importance Weighting

$$J(\theta) = \mathbb{E}_{x \sim P_\theta} \left[f(x) \right]$$

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Finishing Up From Last Lecture



1. Warm up: computing gradient using importance weighting

2. Policy Gradient formulation

Policy Gradient:

Examples of Policy Parameterization (discrete actions)

Recall that we consider parameterized policy $\pi_\theta(\cdot \mid s) \in \Delta(A), \forall s$

1. Softmax Policy for discrete MDPs:

$\theta_{s,a} \in \mathbb{R}, \forall s, a \in S \times A$

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$$f_\theta : S \times A \mapsto \mathbb{R}$$

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In high level, think about π_θ as a classifier which has its parameters to be optimized

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Let's apply the importance weighting trick!

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In class excerise: can you plug in the definition of ρ_θ into the above expression, and see if you can simplify the gradient formulation further

Summary so far for Policy Gradients

We derived the most classic PG formulation:

$$\nabla J(\theta) = \mathbb{E}_{\tau \sim \rho_\theta(\tau)} \left[\sum_{h=0}^{H} \nabla_\theta \ln \pi_\theta(a_h \,|\, s_h) \cdot R(\tau) \right]$$

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Increase the likelihood of on trajectories with high total reward

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Exercise:

Show this simplified version is equivalent to REINFORCE

REINFORCE works surprisingly well on generative models

We train LLMs to summarize reddit post

Example Post

SUBREDDIT: r/dogs
TITLE: [HELP] Not sure how to deal with new people/dogs and my big ole pup
POST: I have a three year old Dober/Pit mix named Romulus ("Rome" for short). I live with 3 other dogs: a 10 year old labrador, a 2 year old French Bulldog and a 8 year old maltese mix. The four of them get along just fine, Rome and the Frenchie are best best best best friends. He isn't the best at meeting new people, but not ALWAYS….Then, the crux of the matter: I want to have a 4th of July party. Several people want to bring their dogs. I doubt I can say "no dogs allowed" and I don't want to let everyone else bring their dog and make mine stay at day care all day.
TL;DR:

Summary:

HOW do I introduce new people? HOW do I introduce new dogs? WHAT do I do about 4th of July??

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Dataset Composition

- 210K Prompts total
  - 117K Prompts with *Human written summarizations*
  - 93K Prompts with *Human Preference Labels*

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Used to pre-train a reward model (details will come soon)

Using RL to optimize 7B size model on TL;DR

Model size	Algorithm	Winrate (↑)
6.9B	SFT	45.2 (±2.49)
	DPO	68.4 (±2.01)
	REINFORCE	70.7*
	PPO	77.6‡
	RLOO ($k = 2$)	74.2*
	RLOO ($k = 4$)	<u>77.9*</u>
	REBEL	**78.1** (±1.74)

* directly obtained from Ahmadian et al. (2024)

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[REBEL: Reinforcement Learning via Regressing Relative Rewards, Neurips 2024]

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Simple REINFORCE already achieves 70% win-rate over humans (evaluated by GPT4)

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Recap: the REINFORCE Algorithm

Initialize θ

While True:

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update: $\theta \Leftarrow \theta + \eta g$ (or adaptive methods like Adam)

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Often require large n to reduce the variance, especially when policy π_θ is quite random

Today's Question:

How to reduce Variance in Policy Gradient?

Outline:

1. A $Q(s, a)$ based Policy Gradient

2. Variance Reduction via A Baseline
(i.e., an $A(s, a)$ based PG)

3. Algorithm: Put everything together

Notations

$$\mathscr{M} = \{P, r, \gamma, \mu, S, A\} \quad \text{where } s_0 \sim \mu$$

$$V^{\pi}(s) = \mathbb{E}\left[\sum_{h=0}^{\infty} \gamma^h r(s_h, a_h) \,|\, s_0 = s, a_h \sim \pi\right]$$

$$\text{Objective: } J(\pi) := \mathbb{E}_{s_0 \sim \mu}\left[V^{\pi}(s_0)\right]$$

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$$d_{\mu}^{\pi}(s, a) = (1-\gamma)\sum_{h=0}^{\infty} \gamma^h \mathbb{P}_h^{\pi}(s, a; \mu)$$

$$A^{\pi}(s, a) = Q^{\pi}(s, a) - V^{\pi}(s, a)$$

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For finite horizon setting, we have:

$$\nabla_\theta J(\pi_\theta) = \sum_{h=0}^{H-1} \mathbb{E}_{s_h,a_h\sim\mathbb{P}_h^{\pi_\theta}} \left[\nabla \ln \pi_\theta(a_h \mid s_h) \cdot Q_h^{\pi_\theta}(s_h,a_h) \right]$$

Outline:



1. A $Q(s, a)$ based Policy Gradient

2. Variance Reduction via A Baseline
(i.e., an $A(s, a)$ based PG)

3. Algorithm: Put everything together

Intuition behind Q-based PG:

$$\nabla_\theta J(\pi_\theta) = \sum_{h=0}^{H-1} \mathbb{E}_{s_h, a_h \sim \mathbb{P}_h^{\pi_\theta}} \left[\nabla_\theta \ln \pi_\theta(a_h \,|\, s_h) \cdot Q_h^{\pi_\theta}(s_h, a_h) \right]$$

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After all, recall PI, we know that $\arg\max_a A^{\pi_\theta}(s, a)$ can work
(subject to knowing A^{π_θ} everywhere)

The Advantage-based PG:

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By a Baseline (proof undoes the importance weighting trick), we have:

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Baseline helps variance reduction (formal proof out of scope)

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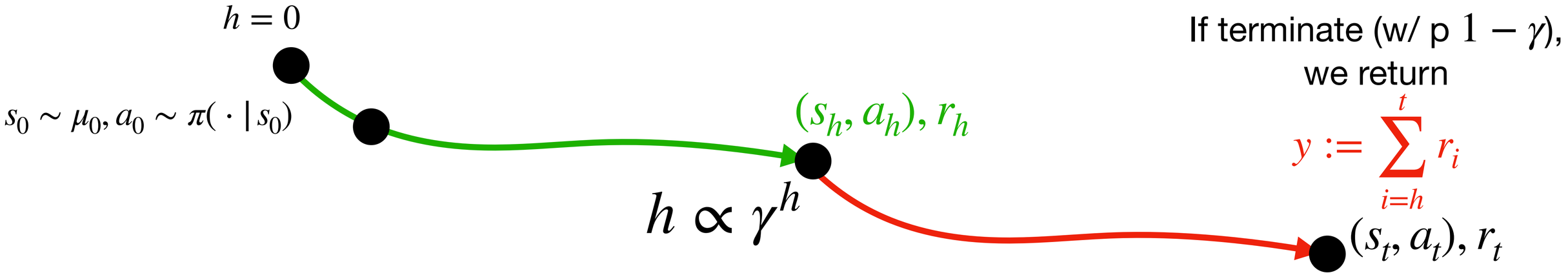
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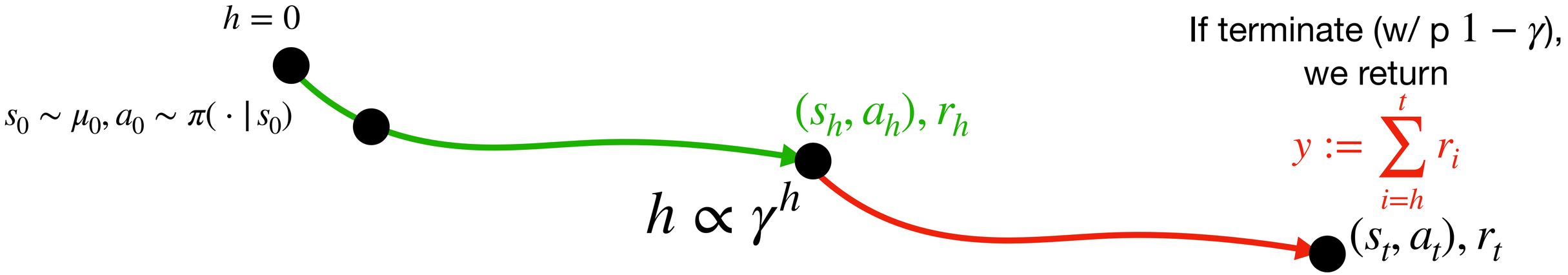
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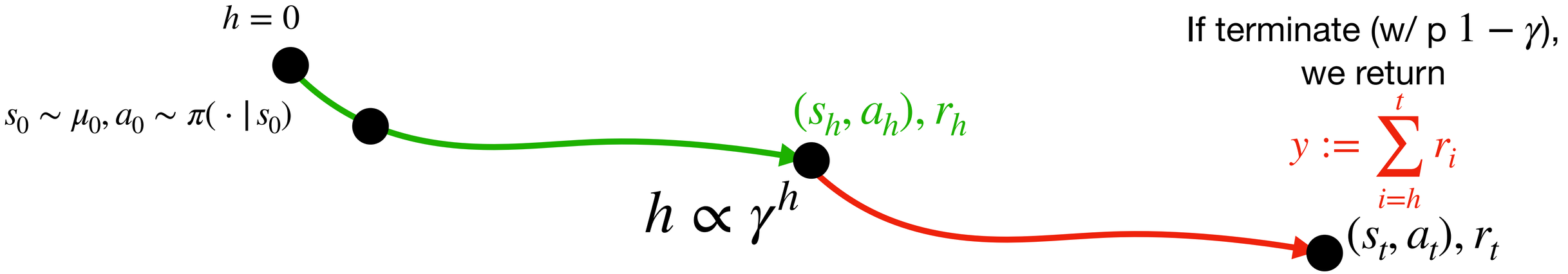


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Algorithm that relies on Stochastic Gradient Ascent

Initialization θ_0

For t = 0, …

Sample $\{s^i, a^i, y^i\}_{i=1}^N$, w/ $s^i, a^i \sim d_\mu^{\pi_{\theta_t}}, \mathbb{E}[y^i] = Q^{\pi_{\theta_t}}(s^i, a^i)$

Form gradient estimate: $g_t = \sum_{i=1}^{N} \nabla_\theta \ln \pi_{\theta_t}(a^i \mid s^i) \cdot y^i / N$

Stochastic GA: $\theta_{t+1} = \theta_t + \eta g_t$

Summary for PG:

Three common PG formulations:

REINFORCE

$$\nabla J(\theta_t) = \mathbb{E}_{\tau \sim \rho_{\theta_t}(\tau)} \left[\left(\sum_{h=0}^{\infty} \nabla_\theta \ln \pi_{\theta_t}(a_h \,|\, s_h) \right) R(\tau) \right]$$

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Three common PG formulations:

REINFORCE

$$\nabla J(\theta_t) = \mathbb{E}_{\tau \sim \rho_{\theta_t}(\tau)} \left[\left(\sum_{h=0}^{\infty} \nabla_\theta \ln \pi_{\theta_t}(a_h \,|\, s_h) \right) R(\tau) \right]$$

PG w/ Q function

$$\nabla_\theta J(\theta_t) = \frac{1}{1-\gamma} \mathbb{E}_{s,a \sim d^{\pi_{\theta_t}}} \left[\nabla_\theta \ln \pi_{\theta_t}(a \,|\, s) \left(Q^{\pi_{\theta_t}}(s,a) \right) \right]$$

Summary for PG:

Three common PG formulations:

REINFORCE

$$\nabla J(\theta_t) = \mathbb{E}_{\tau \sim \rho_{\theta_t}(\tau)} \left[\left(\sum_{h=0}^{\infty} \nabla_\theta \ln \pi_{\theta_t}(a_h \,|\, s_h) \right) R(\tau) \right]$$

PG w/ Q function

$$\nabla_\theta J(\theta_t) = \frac{1}{1-\gamma} \mathbb{E}_{s,a \sim d^{\pi_{\theta_t}}} \left[\nabla_\theta \ln \pi_{\theta_t}(a \,|\, s) \left(Q^{\pi_{\theta_t}}(s,a) \right) \right]$$

PG w/ A function (use $V^\pi(s)$ as a baseline)

$$\nabla_\theta J(\theta_t) = \frac{1}{1-\gamma} \mathbb{E}_{s,a \sim d^{\pi_{\theta_t}}} \left[\nabla_\theta \ln \pi_{\theta_t}(a \,|\, s) \left(A^{\pi_{\theta_t}}(s,a) \right) \right]$$

Feedback and Attendance

https://bit.ly/4gYn2nC

